

Post-Training Large Language Models for Diverse High-Quality Responses

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What does this paper do?

Main goal: post-train LLMs to produce **multiple high-quality and semantically diverse responses**.

- Standard RL post-training improves response quality, but often collapses output diversity.
- This paper proposes **DQO (Diversity Quality Optimization)**.
- DQO adds a **DPP-based diversity objective** on top of standard RL post-training.
- It is implemented on top of **PPO / GRPO**, not as a completely new RL optimizer.

High-level takeaway:

- optimize for **quality** using a reward model
- optimize for **semantic diversity** using the determinant of an embedding similarity matrix

Motivation: Recent Attempts and Limitations

Inference-time methods (temperature scaling, top- k , nucleus sampling):

- Only introduce token-level variation, not semantic diversity.
- Cannot recover modes absent from the learned distribution.

Training-time methods:

- **Token-level entropy regularizer** (Yao et al., 2025): captures lexical but not semantic diversity.
- **Diverse Preference Optimization** (Lanchantin et al., 2025): diversity based on surface features (generation probability, word counts).
- **DPO variant with pairwise distance** (Chung et al., 2025): limited to DPO; pairwise distances can yield degenerate clustering solutions.

Motivation: Our Contributions

Key novelties of DQO:

- **Semantic-level diversity:** operates in embedding space via a pretrained encoder, not at the token level.
- **Determinant-based group diversity:** uses $\det(L)$ instead of pairwise distances, which avoids degenerate clustering and captures linear dependence.
- **Principled quality–diversity trade-off:** reward scales the embedding norm, diversity controls angular spread; governed by a single hyperparameter α .
- **Algorithm-agnostic:** can be combined with PPO, GRPO, or other RL methods.

Preliminary: standard reinforcement learning objective

The standard RL post-training objective is

$$\pi^* = \arg \max_{\pi_{\theta}} \{J(\pi_{\theta}) - \beta KL(\pi_{\theta} \parallel \pi_{\text{ref}})\}.$$

Interpretation:

- $J(\pi_{\theta})$: expected reward of the policy
- $KL(\pi_{\theta} \parallel \pi_{\text{ref}})$: penalty for drifting too far from the reference model
- β : controls the reward–stability tradeoff

What is missing?

- this objective does not explicitly encourage **diversity across multiple sampled outputs**

Preliminary: what is a DPP?

Determinantal Point Process (DPP):

- a probability distribution over **subsets**
- prefers sets whose elements are **non-redundant**
- similar items are less likely to co-occur

Examples:

- search results: avoid returning near-duplicate pages
- summarization: avoid selecting repeated sentences
- LLM outputs: avoid generating many paraphrases of the same answer

Core intuition:

- if one item is selected, a very similar item becomes less likely to also be selected
- therefore DPPs naturally model **diverse subsets**

Preliminary: mathematical expression of a DPP

For a DPP with marginal kernel K ,

$$P(A \subseteq Y) = \det(K_A)$$

where:

- Y : the random subset sampled by the DPP
- A : any subset of candidate items
- K_A : the principal submatrix indexed by A

Special cases:

$$P(i \in Y) = K_{ii}$$

$$P(i, j \in Y) = \begin{vmatrix} K_{ii} & K_{ij} \\ K_{ji} & K_{jj} \end{vmatrix} = K_{ii}K_{jj} - K_{ij}^2$$

Interpretation:

- K_{ii} : marginal inclusion probability of item i
- K_{ij} : similarity / overlap between items i and j
- if K_{ij} is large, then $P(i, j \in Y)$ decreases

Preliminary: explicit 2-item example

Goal of this example: show mathematically that similarity reduces co-occurrence.

Let

$$K = \begin{bmatrix} 0.8 & 0.5 \\ 0.5 & 0.7 \end{bmatrix}$$

Then:

$$P(1 \in Y) = 0.8, \quad P(2 \in Y) = 0.7$$

If the two items were independent, we would expect

$$P(1, 2 \in Y) = 0.8 \times 0.7 = 0.56.$$

But under the DPP,

$$P(1, 2 \in Y) = \begin{vmatrix} 0.8 & 0.5 \\ 0.5 & 0.7 \end{vmatrix} = 0.8 \cdot 0.7 - 0.5^2 = 0.56 - 0.25 = 0.31.$$

What this proves:

- each item is individually likely
- but because they are similar ($K_{12} = 0.5$), they are much less likely to appear together
- this is the basic **repulsion** / **diversity** effect of DPPs

Preliminary: L-ensemble

Definition (L-ensemble)

Let $\mathcal{Y} = \{1, 2, \dots, N\}$ be a ground set, and let $L \in \mathbb{R}^{N \times N}$ be a real symmetric positive semidefinite matrix. An **L-ensemble** defines a distribution over subsets $Y \subseteq \mathcal{Y}$ by

$$P_L(Y = A) \propto \det(L_A).$$

Equivalently,

$$P_L(Y = A) = \frac{\det(L_A)}{\det(L + I)}.$$

Interpretation:

- $\det(L_A)$ is the unnormalized score of subset A
- $\det(L + I)$ is the normalization constant

Preliminary: L -Ensemble — Numerical Example

Example: Let $\phi(y_1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\phi(y_2) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\phi(y_3) = \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix}$.

Gram matrix $L = \Phi^\top \Phi$ where $\Phi = [\phi(y_1), \phi(y_2), \phi(y_3)]$:

$$L = \begin{pmatrix} 1 & 0 & 0.9 \\ 0 & 1 & 0.1 \\ 0.9 & 0.1 & 0.82 \end{pmatrix}$$

- $\det(L_{\{1,2\}}) = 1 \cdot 1 - 0^2 = 1$ (**diverse pair, high prob**)
- $\det(L_{\{1,3\}}) = 1 \cdot 0.82 - 0.9^2 = 0.01$ (**similar pair, low prob**)
- $\det(L_{\{1,2,3\}}) = 0.01$ (adding y_3 contributes little new information)

The DPP strongly favors $\{y_1, y_2\}$ over $\{y_1, y_3\}$ — it **repels** similar items.

Preliminary: relation with Gram matrix

If L is positive semidefinite, then it can be written as

$$L = B^\top B$$

for some matrix B .

If the columns of B are item vectors $b_1, \dots, b_N$, then

$$L_{ij} = b_i^\top b_j.$$

So L is a **Gram matrix**: the matrix of all pairwise dot products.

Why this is important:

- it gives a geometric interpretation of DPPs
- it guarantees L is positive semidefinite

Preliminary: why determinant is volume

$$\det(L_A) = \det(B_A^\top B_A)$$

equals the **squared volume** spanned by the selected vectors.

Geometric picture:

- 2 vectors $\rightarrow$ parallelogram area
- 3 vectors $\rightarrow$ parallelepiped volume
- higher-dimensional vectors $\rightarrow$ high-dimensional volume

Interpretation:

- vectors in very different directions $\rightarrow$ large volume
- redundant / dependent vectors $\rightarrow$ small volume
- linearly dependent vectors $\rightarrow$ zero volume

Preliminary: why determinant is volume

$$P_L(Y) \propto \det(L_Y) = \text{Vol}^2(\{B_i\}_{i \in Y})$$

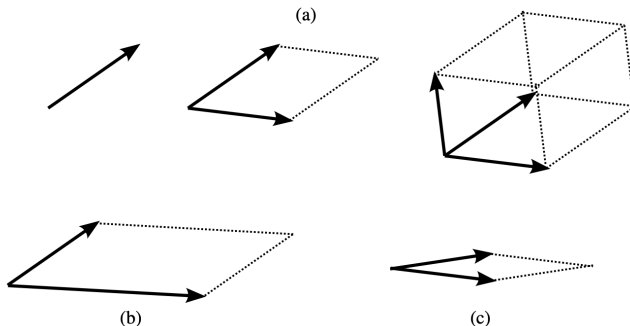


Figure 6: A geometric view of DPPs: each vector corresponds to an element of $\mathcal{Y}$. (a) The probability of a subset Y is the square of the volume spanned by its associated feature vectors. (b) As the magnitude of an item’s feature vector increases, so do the probabilities of sets containing that item. (c) As the similarity between two items increases, the probabilities of sets containing both of them decrease.

DQO: diversity via DPPs

For a prompt x , sample k responses $y_{1:k}$. Define diversity as

$$\text{Div}(y_{1:k}) = \det(L_\phi(y_{1:k}))$$

where

$$L_\phi(y_{1:k})[i, j] = f(\phi(y_i), \phi(y_j)).$$

Components:

- $\phi(y)$: embedding model mapping responses into semantic space
- $f(\cdot, \cdot)$: kernel function, dot product by default
- $\det(L_\phi)$: determinant-based group diversity score

Why this representation?

- similarity is measured in embedding space, so it captures **semantic** rather than lexical diversity
- determinant measures diversity of the **whole set**, not just pairs

Two advantages of DQO diversity

Advantage 1: semantic diversity

- token-level methods only encourage surface variation
- DQO measures diversity in embedding space
- therefore it is closer to what humans mean by “different answers”

Advantage 2: group diversity

- pairwise distance only compares items two at a time
- determinant measures whether the whole set spans many independent directions

Key message:

- DQO optimizes **semantic coverage of the entire response set**

Why pairwise distance can fail

Goal of this example: show that pairwise difference does not imply group diversity.

Take

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

with

$$v_3 = v_1 + v_2.$$

Why pairwise metrics may fail:

- v_1 and v_2 are different
- v_3 is also different from each of them individually
- so pairwise distance may say the set is reasonably diverse

But what is actually true?

- v_3 adds no new direction beyond v_1 and v_2
- the set $\{v_1, v_2, v_3\}$ is linearly dependent
- therefore the spanned volume is 0

What this proves:

- pairwise distance only checks local differences
- determinant checks whether the whole set adds new independent dimensions

Explicit determinant calculation for the 3-vector example

Normalize the vectors:

$$u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Their Gram matrix is

$$G = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

Notice the third row satisfies

$$R_3 = \frac{1}{\sqrt{2}}R_1 + \frac{1}{\sqrt{2}}R_2.$$

Therefore the rows are linearly dependent, so

$$\det(G) = 0.$$

What this proves:

- even though the third vector is not identical to the first two
- it contributes no new independent direction
- determinant correctly says: this set has zero group volume

Quality-diversity objective

DQO optimizes

$$J_{\text{Div}}(\pi_{\theta}) = \mathbb{E}_{x, y_{1:k} \sim \pi_{\theta}(\cdot|x)} \left[\sum_{i=1}^k r(x, y_i) + \alpha \log \det(L_{\phi}(y_{1:k})) - \beta KL(\pi_{\theta} \parallel \pi_{\text{ref}}) \right].$$

Interpretation of each term:

- $\sum_i r(x, y_i)$: response quality
- $\alpha \log \det(L_{\phi})$: diversity bonus
- $\beta KL(\pi_{\theta} \parallel \pi_{\text{ref}})$: regularization to reference model

Why use log?

- determinant can vary over a huge scale
- log makes the diversity term more numerically manageable
- it also turns a multiplicative volume term into an additive objective term

Reward-augmented embedding

Assume $\beta = \alpha$. The paper defines

$$\psi(x, y) = \sqrt{\exp\left(\frac{r(x, y)}{\alpha}\right) \pi_{\text{ref}}(y|x) \phi(y)}.$$

Then the optimal policy satisfies

$$\pi_{\text{div}}(y_{1:k}|x) \propto \det(L_{\psi}(x, y_{1:k})).$$

Interpretation:

- $\phi(y)$: semantic direction of response y
- scalar multiplier: reward + reference-policy preference
- $\psi(x, y)$: reward-augmented semantic embedding

Why can reward-augmented embedding be constructed like this?

Key idea from Appendix B: scaling each embedding vector rescales the determinant.

If the original response embeddings are columns of B , and we scale each column by a_i , then

$$B' = BA, \quad A = \text{diag}(a_1, \dots, a_k)$$

and the Gram matrix becomes

$$L' = ALA.$$

Therefore,

$$\det(L') = \det(A) \det(L) \det(A) = \left(\prod_{i=1}^k a_i^2 \right) \det(L).$$

Why this proves the construction:

- equation (4) contains a product of reward/reference weights across responses
- choosing

$$a_i = \sqrt{\exp(r(x, y_i)/\alpha) \pi_{\text{ref}}(y_i|x)}$$

makes $\prod_i a_i^2$ exactly match those weights

- therefore reward and reference-likelihood can be absorbed into the determinant

Algorithm: practical challenges

Challenge 1: numerical instability

- $\det(L)$ can be very close to 0
- then $\log \det(L)$ can become very large negative
- this destabilizes training

Fix: replace $\log \det(L)$ by

$$\log \det(L + I)$$

- adding identity regularizes the determinant
- makes the diversity term bounded and more stable

Challenge 2: high-variance gradients

- the gradient sums contributions from k sampled responses
- this can create high variance for larger k

Fix: leave-one-out estimator

$$\log \frac{\det(L(y_{1:k}) + I_k)}{\det(L(y_{-i}) + I_{k-1})}$$

- gives each response a leave-one-out diversity contribution
- reduces variance

Lemma

Lemma

Let $\lambda_k \geq \dots \geq \lambda_1$ be the eigenvalues of $L(y_{1:k})$. Then

$$1 + \lambda_k \geq \frac{\det(L(y_{1:k}) + I)}{\det(L(y_{-i}) + I)} \geq 1 + \lambda_1.$$

Since the eigenvalues lie in $[0, k]$, we obtain

$$1 + k \geq \frac{\det(L(y_{1:k}) + I)}{\det(L(y_{-i}) + I)} \geq 1$$

and therefore

$$\log(1 + k) \geq \log \frac{\det(L(y_{1:k}) + I)}{\det(L(y_{-i}) + I)} \geq 0.$$

Why this gives a stable training signal:

- the leave-one-out diversity term is nonnegative
- it is also upper bounded

Implementation

DQO is implemented on top of standard RL post-training.

- **GRPO** for reasoning tasks
- **PPO** for non-reasoning tasks

Pipeline:

- 1 sample k responses for each prompt
- 2 evaluate reward $r(x, y_i)$ for each response
- 3 embed responses with $\phi(y_i)$
- 4 build kernel matrix over the group
- 5 compute the diversity term
- 6 combine reward, diversity, and KL regularization
- 7 update policy using PPO / GRPO

Experiments: city recommendation

Goal: a simple synthetic experiment to visualize whether DQO really improves diversity.

Task:

- prompt the model to recommend a city to visit and give a reason

Compared methods:

- GRPO
- DQO with pairwise-distance diversity
- DQO with determinant-based diversity

Experiments: city recommendation

Main result:

- GRPO repeatedly recommends almost the same city
- pairwise-distance DQO is dominated by a few clusters
- determinant-based DQO produces the broadest coverage of cities



Figure 3: From left to right: GRPO, DQO-pairwise distance, DQO-determinant. Each circle represents a different city, with the size proportional to the number of times it was recommended. For each model, we sampled 100 times with a temperature of 1.0.

Experiments: general tasks

Datasets / tasks:

- **GSM8K**: reasoning
- **CNN/DailyMail**: summarization
- **Dolly**: instruction following
- **CommonGen**: creative writing

Training setup:

- GRPO for GSM8K
- PPO for the other tasks
- reward model provides quality scores
- sentence embedding model provides diversity embeddings

Experiments: metrics

Quality metrics:

- $\text{pass}@n$: best score among n sampled responses
- $\text{pass}@1$: single-response quality
- $\text{pass}@10$: quality when searching over multiple responses

Diversity metrics (computed over 10 responses per prompt):

- **Distinct- n** : ratio of unique n -grams among responses.
- **1 – Self-BLEU**: measures lexical dissimilarity (higher = more diverse).
- **1 – Self-ROUGE**: similar, based on ROUGE scores.
- **LLM-as-a-Judge**: GPT-4o-mini rates diversity (validated against human study).

Experiment: Table 1 — DQO vs. Baselines

Method / Task	Diversity $\uparrow$				Quality $\uparrow$	
	distinct-1	distinct-4	self-bleu	self-rouge	pass@1	pass@10
Dolly						
PPO	0.24	0.64	0.41	0.49	5.65	8.39
GRPO-likelihood	0.26	0.70	0.46	0.54	5.86	8.50
GRPO-entropy	0.36	0.75	0.56	0.57	4.71	7.70
DQO	0.28	0.69	0.46	0.54	5.92	8.74
GSM8K						
GRPO	0.09	0.32	0.09	0.21	76.8	87.9
GRPO-likelihood	0.26	0.86	0.53	0.59	50.9	80.4
GRPO-entropy	0.10	0.38	0.12	0.25	77.0	92.6
DQO	0.10	0.42	0.14	0.31	76.3	91.2

Key points to explain:

- On **Dolly**, DQO improves diversity and also slightly improves quality over PPO.
- On **GSM8K**, DQO improves diversity over GRPO while keeping pass@1 close and improving pass@10.
- Compared with other diversity-aware baselines, DQO is the most balanced on both tasks.

Main message:

- DQO consistently achieves strong **quality + diversity** together.

Experiment: Figure 4 — $\text{pass}@n$ Performance

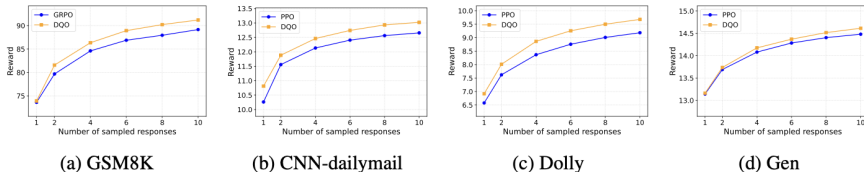


Figure 4: The performance of the trained model on $\text{pass}@n$ metrics. For DQO, we set hyperparameters $\alpha = 1.0$ and $k = 4$.

- x-axis: number of sampled responses
- y-axis: $\text{pass}@n$ quality
- compare baseline reward-only training with DQO

Main observations:

- DQO usually matches or slightly improves $\text{pass}@1$
- DQO improves more clearly when n becomes larger

Interpretation:

- DQO gives a better set of candidate responses to search over

Experiment: Figure 5: diversity metrics

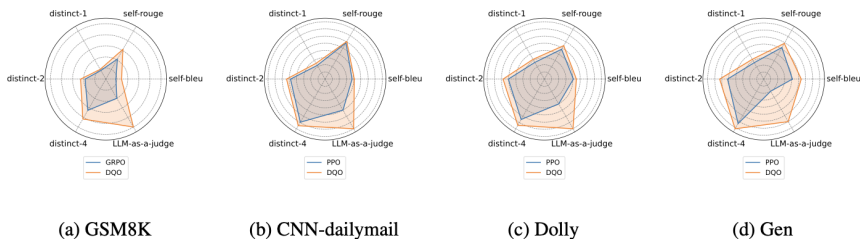


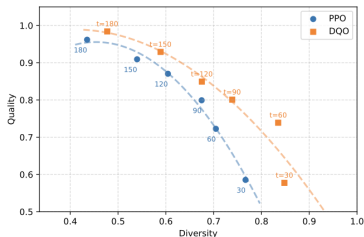
Figure 5: The performance of the trained model on six diversity metrics. For each metric, the higher value means the higher diversity. For DQO, we set hyperparameters $\alpha = 1.0$ and $k = 4$. And the diversity metrics are calculated across 10 generated responses per prompt.

- each radar plot shows six diversity metrics
- compare baseline vs DQO on each task

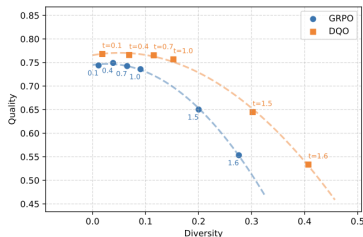
Main observations:

- DQO consistently expands the diversity profile
- gains appear across both lexical and semantic metrics
- LLM-as-a-Judge especially supports semantic diversity improvements

Figure 6: Pareto frontier



(a) Gen



(b) GSM8K

Figure 6: Pareto frontiers on quality and diversity of our model and the baseline. On the left, each point is a model trained with different training steps and the sampling temperature is set as 1.0. On the right, we take the final trained model but vary the sampling temperature.

- x-axis: diversity; quality
- points correspond to different training steps or different temperatures

Main observations:

- DQO lies on a better quality-diversity frontier
- this holds both across training and across inference temperatures

6. Table 2: ablation on α and k

Method	Diversity $\uparrow$				Quality $\uparrow$	
	distinct-1	distinct-4	self-bleu	self-rouge	<i>pass</i> @1	<i>pass</i> @10
PPO	0.24	0.64	0.41	0.49	5.65	8.39
$\alpha = 0.5, k = 4$	0.28	0.69	0.44	0.53	5.84	8.79
$\alpha = 1.0, k = 4$	0.28	0.69	0.47	0.54	5.92	8.73
$\alpha = 1.5, k = 4$	0.33	0.79	0.57	0.61	5.47	8.56
$\alpha = 2.0, k = 4$	0.35	0.82	0.54	0.64	5.42	8.69
$k = 2, \alpha = 1.0$	0.24	0.62	0.40	0.50	5.71	8.13
$k = 4, \alpha = 1.0$	0.28	0.69	0.47	0.54	5.92	8.73
$k = 6, \alpha = 1.0$	0.31	0.76	0.49	0.58	5.71	8.83
$k = 8, \alpha = 1.0$	0.32	0.79	0.52	0.61	5.64	8.64

- α : how much weight to put on diversity
- k : how many responses are sampled per prompt

Main observations:

- larger α usually increases diversity; larger k also usually increases diversity
- too much diversity pressure may slightly hurt quality

Interpretation:

- DQO is fairly robust and does not require extremely delicate tuning

Conclusion

Summary:

- RL post-training improves quality but often collapses diversity
- DQO introduces a determinant-based semantic diversity objective
- the determinant captures **group-level** diversity, beyond pairwise distance
- reward-augmented embeddings unify quality and diversity geometrically
- DQO can be implemented on top of PPO / GRPO

Main conceptual shift:

- from optimizing one good response
- to learning a policy that can generate **multiple high-quality and meaningfully different responses**

Limitation to mention in discussion:

- depends on reward model quality and embedding quality
- outcome reward alone can lead to reward hacking